Mehdi Safaee

Interests

- Computer Vision
- Reinforcement Learning
- Autonomous Robotics
- Cognitive Sciences

About

I am a computer (robotics) vision scientist and engineer with my main concentration on developing robust and efficient solutions for mission-critical autonomous systems.

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Education

- M.Sc. in Computer Science at Simon Fraser University **(2022-2023)**
- **(Optional)** Minor in Electrical Engineering at Amirkabir University of Technology **(2015-2019)**
- B.Sc. in Computer Engineering at Amirkabir University of Technology **(Honors)** **(2013-2019)**
- Diploma of Mathematics at Ghalam High School **(Honors)** **(2009-2013)**

Notable Achievements

- Member of **Iranian National Elites Foundation** **(2021)**
- **Ranked 1st** in a National **Artificial Intelligence (Rahneshan)** Contest **(2021)**
- Ranked Among **Top 10% Students of Computer Engineering Faculty** (~100 Students) **(2013-2019)**
- Ranked Among **Top Three Students** During All Years of High School (~50 students) **(2009-2012)**
- Semi-finalist in Mathematics, Physics, and Astrology Olympiads in High School **(2010-2011)**
- **Ranked 1st** in National **Robotics** Contest (Students Track in Special Path-Finding Robots) **(2010)**
- Silver and Bronze Medals at Iranian Youth Chess Championship (Tehran Province) **(2006-2007)**

Related Outstanding Courses

Course	*Grade*	*Standing*	*Institution*
Neural Networks	*A+*	***Ranked 1st** in our class of ~20 graduate students*	***AUT***
Artificial Intelligence	*A*	***Ranked 1st** in our class of ~30 undergrad students*	***AUT***
Probability and Statistics	*A+*	***Ranked 1st** in our class of ~50 undergrad students*	***AUT***
Fundamentals of Programming	*A+*	***Ranked 2nd** in our class of ~30 undergrad students*	***AUT***
Advanced Programming	*A+*	***Ranked 2nd** in our class of ~50 undergrad students*	***AUT***
Real-Time Embedded Systems	*A+*	***Ranked 1st** in our class of ~30 undergrad students*	***AUT***
Signals and Systems	*A*	***Ranked 2nd** in our class of ~20 undergrad students*	***AUT***
Linear Control System	*A+*	***Ranked 1st** in our class of ~30 undergrad students*	***AUT***

Notable Publications

- Realtime Intelligent Path-finding Agent with DQN Algorithm **(Sep 2019)**
  - *B.Sc. Project - Under the supervision of Dr. Mehdi Sedighi*

Notable Research Experience

- Research Assistant at SFU Visual Computing Group **(Jan 2022 - Current)**
  - *Researching the robustness of computer vision algorithms for autonomous navigation*
  - *Under the supervision of Prof. Ali Mahdavi-Amiri*
- Research Assistant at Amirkabir Computational Intelligence lab **(July 2019 - June 2021)**
  - *Researched various reinforcement learning algorithms and their applications in robotics*
  - *Under the supervision of Prof. M. B. Menhaj*

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Skills and Knowledge

Programming

Python Programming ***[confident]***
C/C++ Programming ***[familiar]***
Basics of Data Structures ***[familiar]***
Software Eng. Principles ***[familiar]***
Linux and Bash Scripting ***[familiar]***

Mathematics

Linear Algebra ***[confident]***
Probabilities and Statistics ***[confident]***
Optimization Basics ***[familiar]***

Artificial Intelligence

Reinforcement Learning ***[familiar]***
Search Algorithms ***[familiar]***
Deep Learning/Pytorch ***[confident]***
Neurobiology ***[confident]***
Machine Learning ***[familiar]***

Robotics

OpenCV ***[confident]***
ROS/ROS2 ***[familiar]***
RaspberryPi ***[confident]***
Jetson Series ***[familiar]***

Other Notable Skills

Latex ***[familiar]***
Parallel Processing (MPI/openMP) ***[familiar]***
HTML/CSS/Javascript ***[familiar]***

Work Experience

- Computer Vision Engineering **Team Lead** at Farand Company **(Jun 2021 - Dec 2021)**
  - *Developed the perception system of the MOOR-III industrial-grade mobile robot*
  - *Developed interfacing and processing solutions for an industrial-grade thermal camera*
- Computer Vision Engineer (Freelance) at Deliware Group **(Oct 2020 - Jun 2021)**
  - *Developed a fully distributed and accelerated perception (Object Det., Segmentation, and Depth Est.) pipeline*
  - *Developed efficient and fast video/data streaming methods for remotely controlling robots*
- Computer Vision Engineering at Farand Company **(Dec 2019 - Jun 2021)**
  - *Developed a high-speed image classifier (120 fps @ 640x480) for industrial automation*
  - *Developed solutions for intrusion detection on embedded systems (using minimal hardware)*
- Signal Processing Engineer at Institute of Cognitive Science Studies **(July 2017 - Sep 2018)**
  - *Gathering, pre-processing, and post-processing of experimental neural data with Matlab*
  - *Providing manual for ARTINIS fNIRS neural data acquisition devices*
- Part-time private programming/robotics/image processing instructor **(Summer 2015 - current)**
  - *Taught fundamentals of Python programming, Raspberry Pi/Arduino interfacing, and Linux scripting*
  - *Taught basics of image processing in Matlab and OpenCV (both Python and C++)*

Notable Teaching Experience

- **Lecturer** for Fundamentals of Linux Scripting workshop **(Farand Group)** **(Fall 2021)**
- **Lecturer** for Robot Operating System (ROS2) workshop **(Deliware Group)** **(Spring 2021)**
- **Lecturer** for Fundamentals of Linux Scripting workshop **(Deliware Group)** **(Fall 2020)**
- **Lecturer** for Fundamentals of Reinforcement Learning workshop **(IPM)** **(Summer 2019)**
- **Lecturer** for Fundamentals of Image Processing workshop **(IPM)** **(Summer 2019)**
- **Head** Teaching assistant of Artificial Intelligence course **(Amirkabir University)** **(Spring 2019)**
- **Head** Teaching assistant of Signals and Systems course **(Amirkabir University)** **(Spring 2019)**
- **Head** Teaching assistant of Microprocessors Design course **(Amirkabir University)** **(Spring 2019)**
- **Lecturer** for Fundamentals of Tensorflow workshop **(Amirkabir University)** **(Fall 2018)**
- Teaching assistant of Data Mining course **(Amirkabir University)** **(Fall 2018)**
- Teaching assistant of Artificial Intelligence course **(Amirkabir University)** **(Fall 2018)**
- Teaching assistant of Signals and Systems course **(Amirkabir University)** **(Fall 2018)**
- Teaching assistant of Microprocessors Design course **(Amirkabir University)** **(Fall 2018)**
- Teaching assistant of Signals and Systems course **(Amirkabir University)** **(Fall 2017)**
- Teaching assistant of Microprocessors Design course **(Amirkabir University)** **(Fall 2017)**
- **Head** Teaching assistant of Electronic Circuits course **(Amirkabir University)** **(Spring 2017)**
- **Head** Teaching assistant of Engineering Mathematics course **(Amirkabir University)** **(Fall 2016)**
- Teaching assistant of Electronic Circuits course **(Amirkabir University)** **(Spring 2016)**
- Teaching assistant of Logic Circuits Design course **(Amirkabir University)** **(Fall 2015)**